the billions of documents, images, or videos on the Web). Embedding models transform each datapoint into a single-vector "embedding", such that *semantically* similar datapoints are transformed into *mathematically* similar vectors. The embeddings are generally compared via the inner-product similarity, enabling efficient retrieval through optimized maximum inner product search (MIPS) algorithms. However, recent advances, particularly the introduction of multi-vector models like ColBERT, have demonstrated significantly improved performance in IR tasks.

Unlike single-vector embeddings, multi-vector models represent each data point with a *set* of embeddings, and leverage more sophisticated similarity functions that can capture richer relationships between datapoints. For example, the popular Chamfer similarity measure used in state-of-the-art multi-vector models captures when the information in one multi-vector embedding is contained within another multi-vector embedding. While this multi-vector approach boosts accuracy and enables retrieving more relevant documents, it introduces substantial computational challenges. In particular, the increased number of embeddings and the complexity of multi-vector similarity scoring make retrieval significantly more expensive.

In "MUVERA: Multi-Vector Retrieval via Fixed Dimensional Encodings", we introduce a novel multi-vector retrieval algorithm designed to bridge the efficiency gap between single- and multi-vector retrieval. We transform multi-vector retrieval into a simpler problem by constructing fixed dimensional encodings (FDEs) of queries and documents, which are single vectors whose inner product approximates multi-vector similarity, thus reducing complex multi-vector retrieval back to single-vector maximum inner product search (MIPS). This new approach allows us to leverage the highly-optimized MIPS algorithms to retrieve an initial set of candidates that can then be re-ranked with the exact multi-vector similarity, thereby enabling efficient multi-vector retrieval without sacrificing accuracy. We have provided an open-source implementation of our FDE construction algorithm on GitHub.

The challenge of multi-vector retrieval

Multi-vector models generate multiple embeddings per query or document, often one embedding per token. One typically calculates the similarity between a query and a document using Chamfer matching, which measures the maximum similarity between each query embedding and the closest document embedding, and then

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document.

While multi-vector representations offer advantages like improved interpretability and generalization, they pose significant retrieval challenges:

- *Increased embedding volume*: Generating embeddings per token drastically increases the number of embeddings to be processed.
- *Complex and compute-intensive similarity scoring*: Chamfer matching is a non-linear operation requiring a matrix product, which is more expensive than a single vector dot-product.
- *Lack of efficient sublinear search methods*: Single-vector retrieval benefits from highly optimized algorithms (e.g., based on space partitioning) that simultaneously achieve high accuracy and sublinear search times, avoiding exhaustive comparisons. The complex nature of multi-vector similarity prevents the direct application of these fast geometric techniques, hindering efficient retrieval at scale.

Unfortunately, traditional single-vector MIPS algorithms cannot be directly applied to multi-vector retrieval — for example, a document might have a token with high similarity to a single query token, but overall, the document might not be very relevant. This problem necessitates more complex and computationally intensive retrieval methods.

MUVERA: A solution with fixed dimensional encodings

MUVERA offers an elegant solution by reducing multi-vector similarity search to single-vector MIPS to make retrieval over complex multi-vector data much faster. Imagine you have a large dataset of "multi-vector sets" (i.e., sets of vectors) where each set describes some datapoint, but searching through each of these sets is slow. MUVERA's trick is to take that whole group of multi-vectors and squeeze them into a single, easier-to-handle vector that we call a *fixed dimensional encoding* (FDE). A key part is that if you compare these simplified FDEs, their comparison closely matches what you'd get if you compared the original, more complex multi-vector sets. This lets us use much quicker search methods designed for single vectors.

Here's a simplified breakdown of how MUVERA works:

1. *FDE generation*: MUVERA employs mappings to convert query and document multi-vector sets into FDEs. These mappings are designed to capture the

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retrieves the most similar document FDEs.

3. *Re-ranking*: The initial candidates retrieved by MIPS are re-ranked using the original Chamfer similarity for improved accuracy.

A key advantage of MUVERA is that the FDE transformation is data-oblivious. This means it doesn't depend on the specific dataset, making it both robust to changes in data distribution and suitable for streaming applications. Additionally, unlike single-vectors produced by a model, FDE's are guaranteed to approximate the true Chamfer similarity to within a specified error. Thus, after the re-ranking stage, MUVERA is guaranteed to find the most similar multi-vector representations.

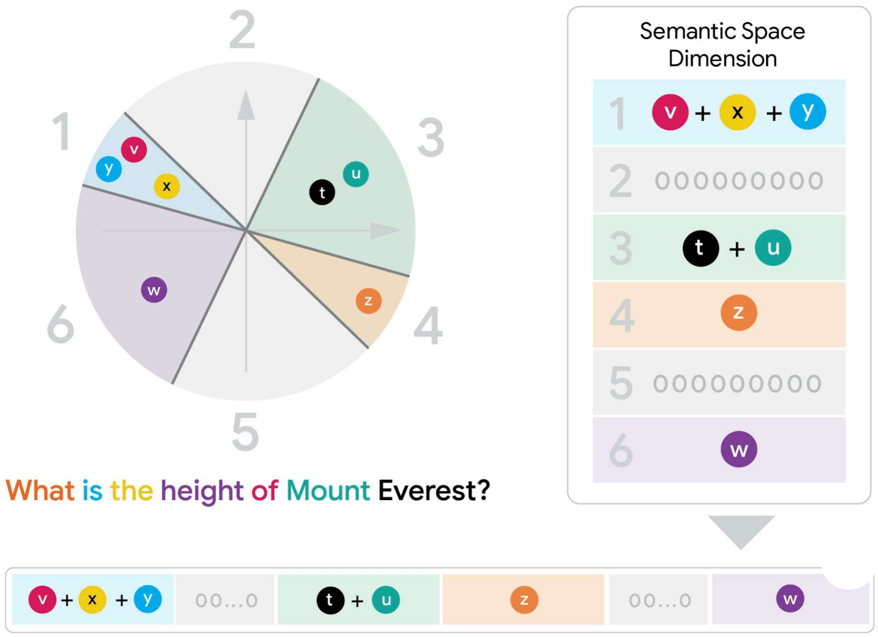

Illustration of the construction of query FDE's. Each token (shown as a word in this example) is mapped to a high-dimensional vector (2-D in the example for simplicity). The high-dimensional space is randomly partitioned by hyperplane cuts. Each piece of space is assigned a block of coordinates in the output FDE, which is set to the sum of the coordinates of the query vectors that land in that piece.